

A representationalist theory of the semantics of reciprocals

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Abstract

As has been noted in the literature, a reciprocal pronoun can have two or more separate antecedents simultaneously, as in *Tom shouted and Mary cried each other's names*, contradicting all previous accounts of the syntax and semantics of reciprocals. This fact and some related observations are shown to motivate a representationalist theory of the syntax-semantics interface in which the semantic representation of each lexical item is a set of elementary predications and semantic composition is allowed to involve taking the union of such sets.

Keywords: right-node raising, the syntax-semantics interface, Head-driven Phrase Structure Grammar, Minimal Recursion Semantics

1 Introduction

It has been noted in the literature that a reciprocal pronoun can have two or more separate antecedents simultaneously, as in (1) and (2). In (1), for instance, the reciprocal pronoun is contained in a right-node-raised phrase, *each other's names*, and has two separate antecedents, *Tom* and *Mary*. A reciprocal pronoun that has more than one antecedent will be referred to as a split-antecedent reciprocal in this article.

- (1) Tom shouted and Mary cried each other's names. (from Chaves (2014))
- (2) John sent a Christmas card, and Mary sent a party invitation, to each other's bosses. (from Kubota and Levine (2020, p. 105))

No currently available theory of reciprocals is capable of assigning appropriate truth conditions to sentences like these.

In fact, split-antecedent reciprocals pose a problem not just for theories of the semantics of reciprocals but for theories of the syntax-semantics interface more generally. It has been recognized that certain types of sentences involving left- or right-node raising can be problematic for the standard theories of the syntax-semantics interface (see Kubota and Levine (2015)), but the problem posed by split-antecedent reciprocals appears to be even more recalcitrant than other problems posed by left- and right-node raising. In sentences like (3a) and (4a), it is at least intuitively obvious what kind of denotation the right-node-raised expression needs to be assigned.

- (3) a. Some woman likes and some man detests every saxophonist. (from Steedman (2000, p. 78))

- b. Some woman likes every saxophonist and some man detests every saxophonist.
- (4) a. John avoided and Bill ignored similar issues. (from Jackendoff (1977, p. 192))
- b. John avoided similar issues and Bill ignored similar issues.

In (3a), the right-node-raised expression, *every saxophonist*, clearly denotes a universal quantifier even when the sentence is given a meaning that cannot be expressed by (3b), and in (4a), the right-node-raised expression, *similar issues*, is clearly an indefinite noun phrase that is being used to refer to some issues even when the sentence is given a meaning that cannot be expressed by (4b). In contrast, in a sentence like (1), it is not easy to say what the right-node-raised expression denotes. Intuitively, the expression seems to mean different things in different conjuncts, in the sense that it seems to refer to Mary's name in relation to the first conjunct and to Tom's name in relation to the second conjunct. It is insufficient, however, to say merely that this right-node-raised expression is assigned different denotations in different conjuncts, since the sentence as a whole is not synonymous with (5), in which the phrase *each other's name(s)* appears twice, without being right-node-raised.

- (5) * Tom shouted each other's name(s) and Mary cried each other's name(s).

In this article, I will argue that an empirically adequate account of split-antecedent reciprocals can be obtained if we adopt a representationalist theory of the syntax-semantics interface in which the semantic representation of each lexical item is a certain type of set and semantic composition is allowed to involve taking the union of such sets. I will start by showing how previous theories fall short in Section 2 and then go on to present the alternative theory.

2 Problems posed by split-antecedent reciprocals

Split-antecedent reciprocals are problematic for all the existing theories of the semantic interpretation of reciprocals, although here I will only discuss the theories presented in Heim, Lasnik, and May (1991b) and Haug and Dalrymple (2020).

The theory presented in Heim, Lasnik, and May (1991b) is based on the view (i) that a reciprocal pronoun always has a plural antecedent, (ii) that there is an unpronounced distribution operator adjoined to that plural antecedent, and (iii) that the reciprocal itself is interpreted as meaning something like "every individual in a given group other than x ," where x is a variable bound by the distribution operator adjoined to the antecedent. In this theory, a sentence like (6) is predicted to mean something like (7).

- (6) Chris and Pat saw each other.
- (7) Each member x of the group consisting of Chris and Pat saw every individual in that group other than x .

Because the theory contains the assumption that part of the meaning of a reciprocal is a variable bound by the distribution operator adjoined to a plural expression, it cannot be applied to a sentence like (1), in which the antecedent of the reciprocal is not a plural expression but two separate singular proper nouns.

Haug and Dalrymple (2020) propose a theory in which (i) a sentence is interpreted under a set of assignments, not just a single assignment, and (ii) a sentence like *Chris and Pat saw each other* is interpreted in such a way that it means "Chris saw Pat" under one assignment and "Pat saw Chris" in another. Since the theory relies on the assumption that a reciprocal pronoun has a single antecedent that denotes different entities under different assignments, it cannot be applied

to a sentence like (1), where the reciprocal does not have a single antecedent but has different antecedents in different conjuncts.

We could envisage a theory that consists of either Heim et al.’s theory or Haug and Dalrymple’s theory and an additional mechanism for dealing with split-antecedent reciprocals, but such a theory would arguably not be a realistic model of speakers’ ability to assign truth conditions to the types of sentences that we are considering. The construction exemplified by (1) and (2) is not something that every native speaker is likely to have encountered in the normal course of language acquisition, and its existence should therefore be a consequence of the way simpler sentences like (6) are analyzed. We need to seek a theory of reciprocals that does not have a component designed to deal specifically with split-antecedent reciprocals.

3 Semantic representations in an alternative account

In presenting an alternative theory, I will first describe what kind of semantic representation each sentence containing a reciprocal is associated with in the theory. Since the proposed semantic representations involve what I will call *reciprocal functions*, I will start by defining that term.¹

- (8) We say that f is a *reciprocal function* when f is a function that satisfies the following condition:
 $\text{dom}(f) \neq \emptyset \wedge \forall x[x \in \text{dom}(f) \rightarrow \exists W \exists Y \exists z[W \subset \text{dom}(f) \wedge x \notin W \wedge W \neq \emptyset \wedge \langle x, W \rangle \in f \wedge x \in Y \wedge \langle z, Y \rangle \in f]]$.

Some examples of reciprocal functions are given in (9).

- (9) a. $\{\langle \text{Chris}, \{\text{Pat}\} \rangle, \langle \text{Pat}, \{\text{Chris}\} \rangle\}$
 b. $\{\langle \text{Alex}, \{\text{Chris}, \text{Pat}\} \rangle, \langle \text{Chris}, \{\text{Pat}, \text{Alex}\} \rangle, \langle \text{Pat}, \{\text{Alex}, \text{Chris}\} \rangle\}$
 c. $\{\langle \text{Alex}, \{\text{Chris}\} \rangle, \langle \text{Chris}, \{\text{Pat}\} \rangle, \langle \text{Pat}, \{\text{Alex}\} \rangle\}$

Each element in a function is an ordered pair consisting of a possible argument of the function and the value that the function returns when it is applied to that argument. Thus, for instance, when f is as shown in (9b), $f(\text{Chris}) = \{\text{Pat}, \text{Alex}\}$.

I submit that a sentence like (6) is associated with a semantic representation like (10), which contains an existentially bound variable, f , which is forced to denote a reciprocal function, as I will explain below.

- (10) $c = \text{Chris} \wedge p = \text{Pat}$
 $\wedge \text{some}(z, z = \{c, p\},$
 $\quad \exists f[\text{each_other}(f,$
 $\quad \quad \text{every}(x, \text{member_of}(x, z),$
 $\quad \quad \quad \text{every}(y, \text{member_of}(y, f(x)),$
 $\quad \quad \quad \text{saw}(x, y)))])$

In this proposed semantic representation, quantification other than that expressed by the existential quantifier binding f is expressed by using a three-place predicate whose first argument is the variable bound by the quantifier, whose second argument is the restrictor, and whose third argument is the nuclear scope. The first two lines of (10) are the contribution from the subject noun phrase *Chris and Pat*, the fourth line is the contribution from the unpronounced distribution operator that is assumed to be adjoined to the verb phrase, the final line is the contribution from

¹The notation “ $\text{dom}(f)$ ” represents the domain of the function f , which is defined as $\{x \mid \exists y[\langle x, y \rangle \in f]\}$.

the verb *saw*, and the third line and the fifth line are the contribution from the reciprocal pronoun *each other*. The predicate *member_of*, used in lines 4 and 5, is assumed to be satisfied if and only if the denotation of its first argument is a member of the set denoted by its second argument. The expression “ $f(x)$ ” in line 5 is meant to denote the value that is returned when the function denoted by f is applied to the denotation of x .

The predicate *each_other*, used in line 3, is interpreted according to the definition given in (11). The term *appropriately minimal*, used in (11), is defined in (12).

(11) The meaning of the *each_other* predicate:

- a. $\llbracket \text{each_other}(f, X) \rrbracket^{M,g}$ is defined and is either 1 or 0 if and only if
 - $g(f)$ is a reciprocal function,
 - f is appropriately minimal for X under g , and
 - X contains no sub-constituent for which f is appropriately minimal under some assignment h such that $h \supseteq g$.
- b. $\llbracket \text{each_other}(f, X) \rrbracket^{M,g} = 1$ if and only if
 - $\llbracket \text{each_other}(f, X) \rrbracket^{M,g}$ is defined and
 - $\llbracket X \rrbracket^{M,g} = 1$.

(12) A variable f is *appropriately minimal* for a formula X under an assignment g if and only if

- $g(f)$ is a function,
- $\llbracket X \rrbracket^{M,g}$ is defined, and
- there is no assignment g' such that
 - $\text{dom}(g') = \text{dom}(g)$,
 - $g'(x) = g(x)$ for every variable x in $\text{dom}(g)$ other than f ,
 - $g'(f)$ is a proper subset of $g(f)$, and
 - $\llbracket X \rrbracket^{M,g'}$ is defined.

The definition in (11) says that *each_other*(f, X) is true only if f denotes a reciprocal function that makes X true. Additionally, the second clause of (11a) says that the function denoted by f must be a minimal function that allows X to have a denotation. And the third clause of (11a) says that, for each sub-constituent of X , the function denoted by f must not be a minimal function that allows that sub-constituent to have a denotation; this clause is necessary in order to prevent the *each_other* predicate from taking excessively wide scope, as will be explained in detail in Section 5 below.

In order for the proposed theory to function properly, it is necessary to specify exactly when the denotation of each type of formula is undefined. Let us assume that $\llbracket \text{member_of}(x, y) \rrbracket^{M,g}$ is undefined if and only if either $\llbracket x \rrbracket^{M,g}$ or $\llbracket y \rrbracket^{M,g}$ is undefined. Let us also assume that $\llbracket \text{every}(x, R, S) \rrbracket^{M,g}$ is undefined if and only if either (i) there is no assignment g' such that $g' \supseteq g$ and $\llbracket R \rrbracket^{M,g'} = 1$ or (ii) for some assignment g' such that $g' \supseteq g$ and $\llbracket R \rrbracket^{M,g'} = 1$, $\llbracket S \rrbracket^{M,g'}$ is undefined.

Let us see what truth conditions the semantic representation in (10) expresses, given these definitions. Since the second argument of the *every* predicate in line 4 is true if and only if the value assigned to the variable x is either Chris or Pat, this *every* predicate is satisfied if and only if its third argument (shown in lines 5 and 6) is true both under an assignment that assigns Chris to x and under an assignment that assigns Pat to x . In order for that third argument to be true under an assignment that assigns Chris to x , the domain of the function denoted by f needs to contain Chris, because otherwise the denotation of this third argument would be undefined. Likewise, in order for that third argument to be true under an assignment that assigns Pat to x , the domain of

the function denoted by f needs to contain Pat. Thus, the domain of the function denoted by f needs to contain both Chris and Pat in order for the *every* predicate in line 4 to be satisfied. Since there is nothing else that the domain of the function denoted by f needs to contain in order for the predicate in question to be satisfied, and since the definition in (11) says that the first argument of the *each_other* predicate has to denote a minimal function that makes the second argument true, it follows that the domain of the function denoted by f needs to contain Chris and Pat and nothing else in order for the *each_other* predicate in line 3 to be satisfied. Because the only reciprocal function whose domain contains Chris and Pat and nothing else is the function shown in (9a) above, and because (in conformity with the third clause in the definition in (11a)²) the second argument of the *each_other* predicate does not contain any sub-constituent that is allowed to have a denotation only if the domain of the function denoted by f contains both Chris and Pat, the foregoing entails that the *each_other* predicate in line 3 is satisfied (and the entire semantic representation is consequently true) if and only if its second argument (shown in lines 4 through 6) is true under an assignment that maps f to $\{\langle \text{Chris}, \{\text{Pat}\} \rangle, \langle \text{Pat}, \{\text{Chris}\} \rangle\}$. Thus, sentence (6) is correctly predicted to be true if and only if Chris saw every member of the set $\{\text{Pat}\}$ and Pat saw every member of the set $\{\text{Chris}\}$.

Following the same strategy of using an existentially bound variable representing a reciprocal function, sentence (1), which contains a split-antecedent reciprocal, can be associated with a semantic representation like (13). Here I am assuming that the noun *names* in this sentence is a dependent plural licensed by the presence of the reciprocal pronoun³ and is hence interpreted in a number-neutral way.

$$\begin{aligned}
 (13) \quad & t = \text{Tom} \wedge m = \text{Mary} \\
 & \wedge \exists f [\text{each_other}(f, \\
 & \quad \text{and}(\text{every}(w, \text{member_of}(w, f(t)), \\
 & \quad \quad \text{some}(x, \text{name_or_names_of}(x, w), \\
 & \quad \quad \quad \text{shouted}(t, x))), \\
 & \quad \text{every}(y, \text{member_of}(y, f(m)), \\
 & \quad \quad \text{some}(z, \text{name_or_names_of}(z, y), \\
 & \quad \quad \quad \text{cried}(m, z)))))]
 \end{aligned}$$

I assume that the *and* predicate, used in line 3, is satisfied if and only if all its arguments are true, and that the denotation of a predication formed by this predicate is undefined if the denotation of one or more of its arguments are undefined. Because the first argument of the *and* predicate here can be true only if the domain of the function denoted by f contains Tom and the second argument of the predicate can be true only if the domain contains Mary, it follows that the *and* predicate in line 3 can be satisfied only if the domain of the function denoted by f contains both Tom and Mary. Given the definition in (11), that means that the *each_other* predicate in line 2 can be satisfied only if the domain of the function denoted by f contains Tom and Mary and nothing else. The second argument of the *each_other* predicate, shown in lines 3 through 8, does not contain any sub-constituent that is allowed to have a denotation only if the domain of the function denoted by f contains both Tom and Mary, so the third clause of (11a) is satisfied when the value assigned to f is a function whose domain is $\{\text{Tom}, \text{Mary}\}$. Therefore the semantic representation in (13) is

²This third clause is unnecessary as far as the current example and the other examples discussed in this section are concerned. Its *raison d'être* will be explained in Section 5 below, as stated earlier.

³The reciprocal pronoun *each other* is grammatically plural, as can be seen from the fact that a pronominal that is coindexed with it must be plural in number, as in *John and Mary told each other that they should leave*, which can be interpreted as saying “John told Mary that *she* should leave and Mary told John that *he* should leave” (see Heim, Lasnik, and May (1991a)).

true if and only if an assignment that maps f to the function $\{\langle \text{Tom}, \{\text{Mary}\} \rangle, \langle \text{Mary}, \{\text{Tom}\} \rangle\}$ (i.e. the only reciprocal function whose domain is $\{\text{Tom}, \text{Mary}\}$) makes the second argument of the *each_other* predicate true. That means that sentence (1) is correctly predicted to be true if and only if Tom shouted the name or names of every member of the set $\{\text{Mary}\}$ and Mary cried the name or names of every member of the set $\{\text{Tom}\}$.

The need to have an existential quantifier that binds a variable representing a reciprocal function becomes clearer when we consider an example like (14), which involves disjunction.

(14) Chris shouted or Pat cried, and Mary whispered, each other's names.

In the proposed account, this sentence is associated with the semantic representation shown in (15).

(15) $c = \text{Chris} \wedge p = \text{Pat} \wedge m = \text{Mary}$
 $\wedge \exists f [\text{each_other}(f,$
 $\quad \text{and}(\text{or}(\text{every}(u, \text{member_of}(u, f(c)),$
 $\quad \quad \text{some}(v, \text{name_or_names_of}(v, u),$
 $\quad \quad \text{shouted}(c, v))),$
 $\quad \text{every}(w, \text{member_of}(w, f(p)),$
 $\quad \quad \text{some}(x, \text{name_or_names_of}(x, w),$
 $\quad \quad \text{cried}(p, x))),$
 $\quad \text{every}(y, \text{member_of}(y, f(m)),$
 $\quad \quad \text{some}(z, \text{name_or_names_of}(z, y),$
 $\quad \quad \text{whispered}(m, z))))]$

Let us assume that $\llbracket \text{or}(d_1, \dots, d_n) \rrbracket^{M,g} = 1$ if and only if $\llbracket d_i \rrbracket^{M,g} = 1$ for some i , irrespective of whether $\llbracket d_j \rrbracket^{M,g}$ is undefined for some j , and see what the semantic representation above means given that assumption. In order for the *each_other* predicate in line 2 to be satisfied, its first argument, f , must be mapped to a reciprocal function that makes the second argument of the predicate true. In order for this second argument to be true, both conjuncts that constitute it must be true. In order for the second conjunct to be true, the domain of the function denoted by f needs to contain Mary. And in order for the first conjunct to be true, the domain of the function denoted by f needs to contain either Chris or Pat; the first disjunct inside the first conjunct is allowed to have a denotation and can be true when the domain of the function denoted by f contains Chris, and the second disjunct inside the first conjunct is allowed to have a denotation and can be true when the domain of the function denoted by f contains Pat. Thus, the *each_other* predicate in line 2 can be satisfied only if the variable f is mapped to a reciprocal function whose domain is either $\{\text{Chris}, \text{Mary}\}$ or $\{\text{Pat}, \text{Mary}\}$, that is, only if the variable f is mapped either to (16) or to (17). Note that the domain of the function denoted by f cannot contain more than two individuals; since the second argument of the *each_other* predicate here can be true under an assignment that maps f to a function whose domain contains only two elements, f will not be appropriately minimal for that argument in the sense defined in (12) if it is mapped to a function whose domain contains more than two elements.

(16) $\{\langle \text{Chris}, \{\text{Mary}\} \rangle, \langle \text{Mary}, \{\text{Chris}\} \rangle\}$

(17) $\{\langle \text{Pat}, \{\text{Mary}\} \rangle, \langle \text{Mary}, \{\text{Pat}\} \rangle\}$.

The second argument of the *each_other* predicate does not contain any sub-constituent that is allowed to have a denotation only if the domain of the function denoted by f contains two or more entities, so the third clause of (11a) is satisfied as well when f denotes either (16) or (17).

Therefore the semantic representation as a whole is true if and only if either an assignment that maps f to (16) or an assignment that maps f to (17) makes the second argument of the *each_other* predicate true. Sentence (14) is thus predicted to be true if and only if either Chris shouted Mary’s name and Mary whispered Chris’s name or Pat cried Mary’s name and Mary whispered Pat’s name, which appears to be a correct prediction. Notice that this prediction cannot be obtained if the variable representing a reciprocal function is not existentially bound and instead given a particular value by the context.

4 The syntax-semantics interface

I now turn to the question of how the kinds of semantic representations I proposed in the previous section can be associated with the corresponding English sentences.

In the theory being proposed, the meaning of reciprocity expressed in a simple sentence manifests itself in two separate places in the semantic representation; in the case of (10), for instance, it is the material in line 3 and the material in line 5 that are responsible for the reciprocal meaning. Let us henceforth refer to the *each_other* predicate, its first argument, and the existential quantifier that binds it, which are in line 3 in (10), as the upper portion of reciprocal meaning, and to the universal quantifier and its first two arguments, which are in line 5 in (10), as the lower portion of reciprocal meaning.

It seems natural to presume that both the upper portion and the lower portion of reciprocal meaning are part of the meaning of a reciprocal pronoun, but at first blush, that presumption seems to lead to a problem when we consider split-antecedent reciprocals. In the example (1), the reciprocal pronoun is shared by the two conjuncts, and has to be given a different interpretation in each conjunct, since intuitively the reciprocal inside the object of the verb *shouted* in the first conjunct has to refer to Mary whereas the reciprocal inside the object of the verb *cried* in the second conjunct has to refer to Tom. Adopting the analysis of right-node raising proposed in Yatabe and Tam (2021), we could allow the reciprocal pronoun to receive different interpretations in different conjuncts and to retain those interpretations even after it has been right-node-raised. However, such an analysis will yield a semantic representation that contains two instances of the upper portion of reciprocal meaning as well as two instances of the lower portion. This is a problem because in the proposed account, the semantic representation for (1) contains two instances of the lower portion of reciprocal meaning but only *one* instance of the upper portion, as shown in (13).

The fact that (13) contains only one instance of the upper portion of reciprocal meaning might seem to indicate that the upper portion here is contributed to the semantic representation not by the reciprocal pronouns (i.e. the one in the first conjunct and the one in the second conjunct, which are fused into one reciprocal pronoun by the operation of right-node raising) but by an unpronounced item that binds both the reciprocals, but such an analysis would be a mistake. In an analysis in which a reciprocal pronoun is assumed to be always bound by a unpronounced item that contributes the upper portion of reciprocal meaning and that item is assumed to be capable of binding more than one reciprocal pronoun at the same time, an example like (18) is incorrectly predicted to be a possible sentence synonymous with (1).

(18) *Tom shouted each other’s name(s) and Mary cried each other’s name(s).

It will be my contention in this section that there are certain independently motivated modifications to be made to the analysis proposed in Yatabe and Tam (2021) and that everything falls into place once those modifications are made.

4.1 The basics of Minimal Recursion Semantics

Before describing those modifications, I will first outline in this subsection the fundamentals of Minimal Recursion Semantics (MRS), which forms the basis of Yatabe and Tam’s theory of right-node raising. In MRS, a semantic representation is always a list of one or more elementary predications⁴; the meaning of a lexical item is represented by a list of elementary predications, and when a phrase is made up of n lexical items whose meanings are represented by the lists of elementary predications $L_1, \dots, L_n$ respectively, the meaning of that phrase is represented by $L_1 \oplus \dots \oplus L_n$, where the symbol $\oplus$ represents list concatenation. An elementary predication consists of a predicate and all its arguments, and each elementary predication is labeled by a handle, which serves to indicate how that elementary predication is to be linked to the other elementary predications. Let us take the following sequence of three elementary predications as our first example.

$$(19) \quad \langle h0 : \text{identical}(x, \text{Chris}), h0 : \text{speaks}(x, y), h0 : \text{identical}(y, \text{Russian}) \rangle$$

These three elementary predications, each prefaced with its handle, namely $h0$, represent the meaning of the words *Chris*, *speaks*, and *Russian* respectively, and the sequence as a whole is meant to represent the meaning of the sentence *Chris speaks Russian*. An elementary predication of the form “ $\text{identical}(i, j)$ ” is assumed to be true if and only if the denotation of the first argument is identical to that of the second argument. An elementary predication of the form “ $\text{speaks}(x, y)$ ” is assumed to be true if and only if the denotation of x speaks the denotation of y . In an MRS representation, elementary predications that are prefaced by the same handle are interpreted as being conjoined with each other. The variables that are not bound by any quantifier are interpreted as bound by an unexpressed existential quantifier at the global top level. Therefore the MRS representation in (19) as a whole is true if and only if there are entities x and y such that x is Chris, y is the Russian language, and x speaks y .

When MRS is integrated into HPSG as it is in Yatabe and Tam (2021), each elementary predication is expressed as a feature structure. For instance, the representation in (19) is expressed as in (20) in such a setup.

$$(20) \quad \left\langle \begin{bmatrix} \text{HNDL} & h0 \\ \text{RELN} & \text{identical} \\ \text{ARG1} & x \\ \text{ARG2} & \text{Chris} \end{bmatrix}, \begin{bmatrix} \text{HNDL} & h0 \\ \text{RELN} & \text{speaks} \\ \text{SUBJECT} & x \\ \text{OBJECT} & y \end{bmatrix}, \begin{bmatrix} \text{HNDL} & h0 \\ \text{RELN} & \text{identical} \\ \text{ARG1} & y \\ \text{ARG2} & \text{Russian} \end{bmatrix} \right\rangle$$

Within each elementary predication, the HNDL value represents the handle that serves as the label of that elementary predication, and the RELN value represents the relation or the predicate that holds together that elementary predication.

Consider next the MRS representation in (21), which illustrates how quantifiers are handled.

$$(21) \quad \langle h0 : \text{every}(x, h1, h2), h1 : \text{smart}(x), h1 : \text{student}(x), h3 : \text{probably}(h4), h5 : \text{agrees}(x) \rangle$$

The list of five elementary predications in (21) is meant to represent the two possible readings of the sentence *Every smart student probably agrees*, one in which *every smart student* outscopes *probably* and another in which the universal quantifier is outscoped by the adverb. In accordance with the notational convention that I have already been using, the three arguments of the predicate

⁴Here I am describing the revised version of MRS that is used in Yatabe and Tam (2021). In the original version of MRS presented in Copestake et al. (2005), the meaning of a linguistic expression is modeled as a bag of elementary predications rather than a list of elementary predications.

every are the variable that the quantifier binds, the restrictor of the quantifier, and the nuclear scope of the quantifier, respectively. An elementary predication of the form “ $\text{every}(x, S, T)$ ” is assumed to be true if and only if every possible value of x that makes S true makes T true as well. The predicates *smart*, *student*, and *agrees* are interpreted in the expected way. An elementary predication of the form “ $\text{probably}(S)$ ” is assumed to be true if and only if the formula S is probably true.

When the handle of a set of elementary predications is identical to the handle filling an argument slot of a predicate, the set of elementary predications is interpreted as filling that argument slot. Thus, the second and the third elementary predication in this example, which both have the handle $h1$, are interpreted as filling the second argument slot of *every*, the argument slot that represents the restrictor of the quantifier.

An MRS representation can serve as an underspecified representation that stands for more than one semantic interpretation because it is assumed that even different handles (such as $h1$ and $h2$) can optionally be interpreted as pointing to the same location in the semantic representation. Let us use the notation $h \approx h'$ to indicate that two handles h and h' point to the same location in the semantic representation, and let us say that h and h' are *locationally equivalent* when $h \approx h'$ holds. I will use example (21) to illustrate what happens when some of the handles in an MRS representation are locationally equivalent. Suppose that $h2 \approx h3$ and $h4 \approx h5$ in (21). Then the elementary predication “ $\text{probably}(h4)$ ” becomes the nuclear scope of *every* and the elementary predication “ $\text{agrees}(x)$ ” becomes the sole argument of *probably*, so the representation in (21) as a whole becomes equivalent to (22).

(22) $\text{every}(x, \text{smart}(x) \wedge \text{student}(x), \text{probably}(\text{agrees}(x)))$

On the other hand, the representation in (21) will become equivalent to (23) if $h4 \approx h0$ and $h2 \approx h5$.

(23) $\text{probably}(\text{every}(x, \text{smart}(x) \wedge \text{student}(x), \text{agrees}(x)))$

The former is the reading in which the universal quantifier takes wide scope over the adverbial, and the latter is the reading in which the adverbial takes scope over the universal quantifier.

A handle h is said to *immediately outscope* a handle h' if and only if h or a handle locationally equivalent to h is attached to an elementary predication that has an argument slot filled by h' or a handle locationally equivalent to h' . We say that h *outscores* h' if and only if h either immediately outscores h' or immediately outscores a handle that outscores h' .

The two readings shown in (22) and (23) are the only readings expressed by the underspecified representation in (21) because it is assumed (i) that there is a global top handle that outscores every other handle that is not locationally equivalent to it, (ii) that, for each handle filling an argument slot of a predicate, there must be at least one elementary predication whose handle is locationally equivalent to it, (iii) that an elementary predication cannot be interpreted as filling more than one argument slot, and (iv) that a handle cannot outscope itself. These assumptions, which require that elementary predications in an MRS representation should form a tree, prevent $h5$ from being locationally equivalent to $h1$ and to $h2$ at the same time in (21), for example. Another constraint that is imposed on MRS representations is that a variable that is bound by a quantifier in an MRS representation cannot be reused as a free variable or as a variable bound by another quantifier in the same MRS representation, although this condition, which is referred to as the variable-binding condition in the literature, plays no role in the above example.

4.2 Modifications to Yatabe and Tam’s theory

I propose two modifications to the MRS-based theory advocated in Yatabe and Tam (2021). First, I submit that the meaning of a lexical item is represented as a set of elementary predications rather than as a list of elementary predications and that the meaning of a sentence is represented as a list of such sets. Given this modification, the meaning of the sentence *Every smart student probably agrees* will be represented as in (24), not as in (21).

$$(24) \quad \langle \{h0 : \text{every}(x, h1, h2)\}, \{h1 : \text{smart}(x)\}, \{h1 : \text{student}(x)\}, \{h3 : \text{probably}(h4)\}, \{h5 : \text{agrees}(x)\} \rangle$$

This is arguably a natural modification to the theory because when the meaning of a lexical item is represented by multiple elementary predications, there is no reason to give any particular linear order to those elementary predications.

The second modification concerns the way that sentences involving right- or left-node raising are interpreted. In Yatabe and Tam (2021), it is assumed that when a lexical item is left- or right-node-raised out of two or more phrases possibly together with some other lexical items and retains the multiple meanings that were assigned to it inside those phrases, the meaning of that left- or right-node-raised lexical item is the concatenation of those multiple meanings, each of which is a list of elementary predications. I propose to discard this assumption and to hypothesize instead that when a lexical item is left- or right-node-raised out of two or more phrases possibly together with some other lexical items and retains the multiple meanings that were assigned to it inside those phrases, the meaning of that left- or right-node-raised lexical item is the union of those multiple meanings, each of which is a set of elementary predications.

These two modifications allow the theory to associate sentences containing split-antecedent reciprocals with appropriate semantic representations. In the case of example (1), in which a reciprocal pronoun has been right-node-raised together with some other lexical items out of two conjuncts, the meaning of the reciprocal pronoun can now be the union of two sets, both containing the upper portion and the lower portion of reciprocal meaning, and will contain two instances of the lower portion and one instance of the upper portion if the upper portions coming from the two conjuncts happen to be identical to each other while the lower portions coming from the two conjuncts happen not to be identical to each other. The resulting semantic representation will be what is shown in (13).

4.3 Lexical entries

Let me next present some lexical entries that will allow the grammar to actually produce the kinds of semantic representations that I have been arguing for. Note that the lexical entries that will be proposed here have some unusual features because they are meant to be integrated into the theory advocated in Yatabe and Tam (2021), which deviates from the textbook version of HPSG (see Pollard and Sag (1994)) in several respects. First, in the theory in question, which is a type of theory that is referred to as Linearization-based HPSG in the literature, each node in a syntactic phrase-structure tree is assumed to have the DOM feature, whose value is what is called an order domain, which represents the prosodic constituency of the morphemes dominated by that node. An order domain is a list of domain objects, each of which represents a prosodic constituent. And second, in this theory, it is the prosodic constituents of sentences rather than their morphosyntactic constituents that are assigned semantic interpretation. The semantic content of each domain object (which, by assumption, is a prosodic constituent) is given as the value of its SYNSEM|CONT|EP feature.

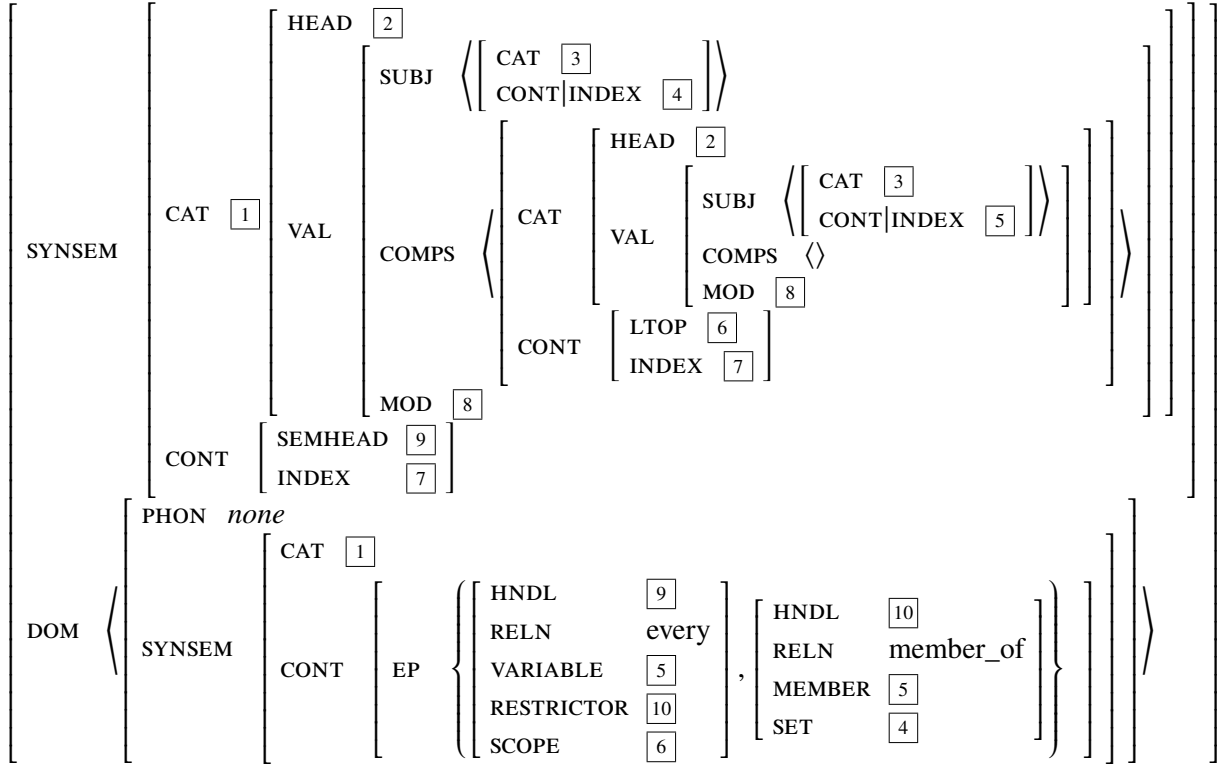


Figure 1: The lexical entry for a silent distribution operator that combines with subject-seeking expressions

I will start with postulating two phonologically null distribution operators in the lexicon of English: one that combines with subject-seeking constituents such as VPs and predicative PPs and another one that combines with DPs. These unpronounced operators give immediately preceding DPs distributive interpretation as opposed to collective interpretation.

The lexical entry shown in Fig. 1 is the first silent distribution operator that I propose. In a sentence like *Three students sang*, this silent distribution operator could be adjoined to the VP, as in (25a), where the symbol DIST1 is meant to stand for the operator.

- (25) a. $[_S \text{ Three students } [_{VP} \text{ DIST1 } [_{VP} \text{ sang}]]]$
b. $\text{some}(X, \text{three}(X) \wedge \text{students}(X), \text{every}(z, \text{member_of}(z, X), \text{sang}(z)))$

The lexical entry shown in Fig. 1 requires (i) that the HEAD value of the lexical item should be identical to the HEAD value of its complement, (ii) that the complement of the lexical item should be a type of expression that needs to combine with a subject to be saturated, (iii) that the subject of the lexical item should have all the grammatical properties that the unexpressed subject of its complement is expected to have, and (iv) that the unexpressed subject of the complement of the lexical item and the subject of the lexical item itself should nevertheless have distinct indices and therefore contribute distinct variables to the semantic representation. In the example in (25a), the silent distribution operator takes as its complement a finite VP, and the constituent made up of the operator and its complement thus becomes a finite VP itself, and is then combined with the subject *three students*. Assuming that this subject is a DP headed by a phonologically null quantificational determiner whose denotation is roughly the same as that of *some*, the sentence (25a) as whole is associated with an MRS representation that is equivalent to the linearly formatted formula shown in (25b), where the argument of the predicate *sang* is z , a variable distinct from the variable contributed by the subject DP, i.e. X . The variable-binding condition, which prohibits a

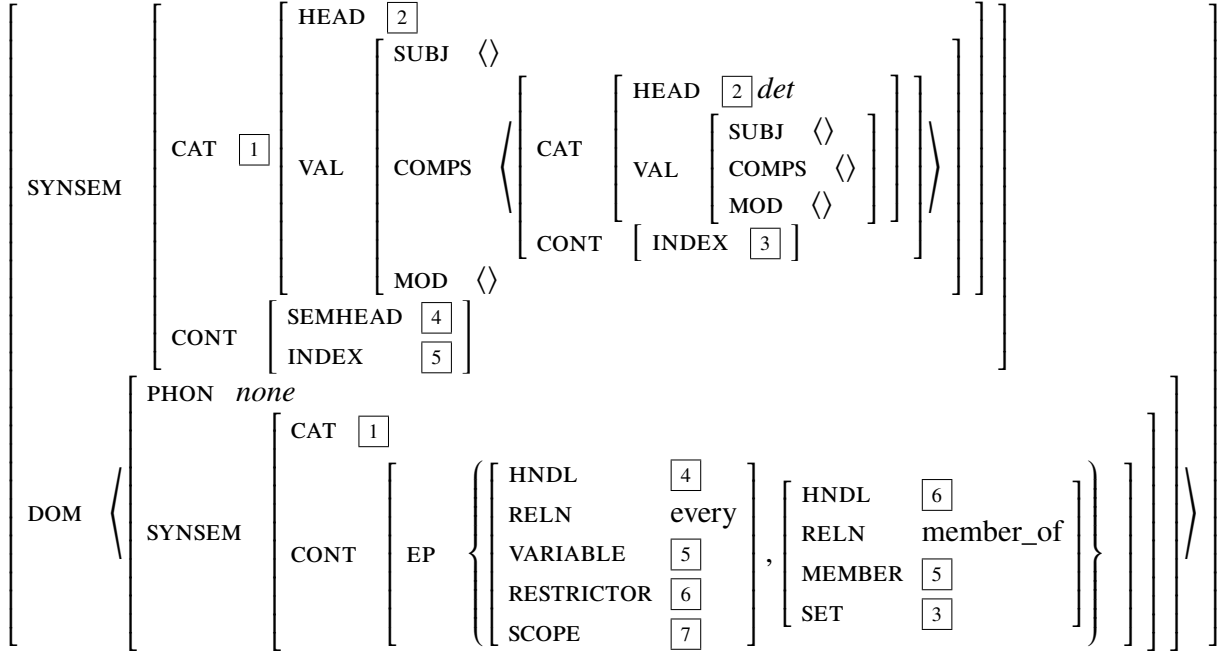


Figure 2: The lexical entry for a silent distribution operator that combines with DPs

variable bound by a quantifier from being reused either as a free variable or as a variable bound by another quantifier in the same MRS representation, guarantees that the existential quantifier contributed by the subject takes wide scope over the universal quantifier contributed by the silent distribution operator, since the restrictor of the latter contains a variable that is to be bound by the former.

The lexical entry shown in Fig. 2 is the second silent distribution operator that I propose. This second operator is used as in (26), where the symbol *DIST2* is meant to stand for the operator.

(26) [_S [_{DP} [_{DP} three students] *DIST2*] [_{VP} sang]]

This syntactic structure also gives rise to the interpretation shown in (25b) and no other interpretation, again thanks to the variable-binding condition.

Postulation of this additional operator is necessitated by sentences like *I questioned them about each other* (Heim, Lasnik, and May (1991a)), where the antecedent of a reciprocal is not a grammatical subject. Postulation of the first silent distribution operator, on the other hand, is necessitated by an example like *people with no love for each other*, in which the antecedent of the reciprocal pronoun is arguably not an overt DP but the unexpressed subject of the predicative PP, which is coindexed with the preceding plural NP.⁵

Let us move on to the question of what kind of lexical entry a reciprocal pronoun needs to be given. My proposal is that *each other* should be given the lexical entry shown in Figure 3 and be associated with the conditions stated in (27). In Figure 3, as in Figures 1 and 2, the set that represents the meaning of the reciprocal pronoun is given as the value of the *EP* feature

⁵There are in fact two known problems with the analysis of the distributive interpretation described in the text. One problem, discussed in Sternefeld (1998), Sauerland (1998), and Yatabe (2023), concerns the availability of the so-called cumulative interpretation in a sentence like *They wrote these six letters to each other*. Another problem, discussed in Williams (1991) and Heim, Lasnik, and May (1991b), concerns the fact that the embedded subject can take scope over the matrix subject in a sentence like *Someone or other has said that each of the men likes the other* but not in a sentence like *Someone or other has thought that they like each other*. I will not go into the details of these problems here because these are problems with the analysis of distribution operators, not problems with the analysis of reciprocals, which the present article is primarily concerned with.

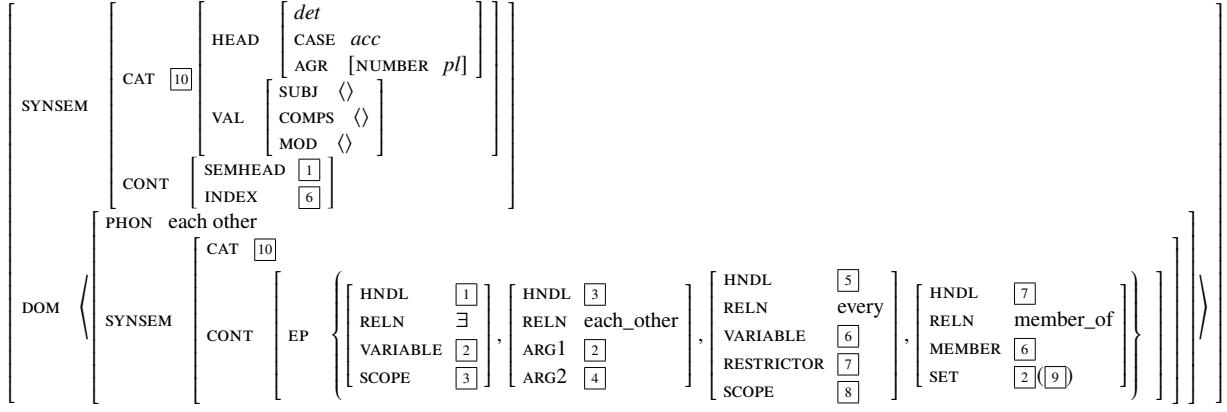


Figure 3: The lexical entry for *each other*

inside the value of the DOM feature; the first two elementary predications in the set represent the upper portion and the other two elementary predications represent the lower portion of reciprocal meaning.

- (27) Conditions: (Note that the denotation of a boxed integer is a piece of feature structure, not something in the outside world.)
- The denotation of $\boxed{9}$ must not be identical to the denotation of $\boxed{6}$.
 - The denotation of $\boxed{8}$ must either outscope or be locationally equivalent to the handle of an elementary predication that contains an occurrence of the variable denoted by $\boxed{9}$.

As will be explained below, the conditions in (27) capture, among other things, the fact that *Tom* and *Mary* are obligatorily chosen as the antecedents of *each other* in (1) despite the reciprocal being an exempt anaphor here in the sense defined in Pollard and Sag (1994).

I note at this point that sentences like the following are beyond the scope of this article.

- The plates are stacked on top of each other.
- You put these two bowls inside each other.

Following Beck (2001) and Haug and Dalrymple (2020), I assume that the phrase *each other* in examples of this type is in fact distinct from the normal reciprocal pronoun, for which the lexical entry in Figure 3 is intended.

4.4 The binding condition for reciprocals

I assume that reciprocal pronouns are subject to a binding condition similar to the version of Binding Condition A that is proposed for reflexive pronouns in Yatabe (2022). That binding condition, which is assumed to apply to semantic representations rather than to syntactic structures (for reasons explained in Yatabe (2022)) but is effectively equivalent to what is proposed in Pollard and Sag (1994), is given in (32). (30) and (31) provide definitions for some of the terms used in (32).

- (30) Definition of *outranking*:

Let E be an elementary predication and let x and y be variables contained in E . We say that x *outranks* y in E if and only if (i) x is the SUBJECT value or the POSSESSOR value of E and y is not, or (ii) x is the OBJECT value of E and y is not the SUBJECT value, the POSSESSOR

value, or the OBJECT value of E , or (iii) x is the SECONDARY-OBJECT value of E and y is not the SUBJECT value, the POSSESSOR value, the OBJECT value, or the SECONDARY-OBJECT value of E .

(31) Definition of an *exempt anaphor*:

An elementary predication E whose RELN value is *member_of*, whose MEMBER value is a variable x , and whose SET value is a parenthesized variable preceded by another variable is an *exempt anaphor* in a semantic representation M if M does not contain an elementary predication in which some variable outranks x .

(32) The binding condition for reciprocals:

Let E be an elementary predication whose RELN value is *member_of*, whose MEMBER value is a variable x , and whose SET value is $f(y)$, i.e. a parenthesized variable y preceded by another variable f . Let M be the semantic representation that contains E . And suppose that E is not an exempt anaphor in M . Then M must contain an elementary predication in which y outranks x .

This binding condition correctly rules out an example like the following.

(33) * They think that John likes each other.

The reciprocal pronoun in (33) has to have a local antecedent because the semantic representation for this sentence contains an elementary predication (contributed by the verb *likes*) whose OBJECT value is a variable that serves as the MEMBER value of one of the elementary predications contributed by the reciprocal pronoun; the presence of that elementary predication prevents (one of the elementary predications contributed by) the reciprocal pronoun from counting as an exempt anaphor in the sense defined in (31).

4.5 Two simple examples

Let me now illustrate how the theory as a whole works, using simple examples. First, let us consider the sentence in (34), which can have the syntactic structure shown in (35), where “dist1” is a silent distribution operator that was introduced above.

(34) The cats saw each other.

(35) [_S [_{DP} The cats] [_{VP} dist1 [_{VP} saw each other]]]

One possible semantic representation that the proposed grammar can assign to this syntactic structure is shown in Figure 4. This semantic representation is a list consisting of five sets, which represent the meaning of *the*, *cats*, *dist1*, *saw*, and *each other* respectively. The VARIABLE value of the sole elementary predication in the first set and the INSTANCE value of the sole elementary predication in the second set are required to be identical by the lexical entries for *the* and *cat* and the standard components of HPSG. The VARIABLE value of the first elementary predication in the third set, the MEMBER value of the second elementary predication in the third set, and the SUBJECT value of the sole elementary predication in the fourth set are required to be identical to each other primarily by the lexical entry for *dist1*, shown in Figure 1 above. The SET value of the second elementary predication in the third set and the VARIABLE value of the sole elementary predication in the first set are also required to be identical to each other primarily by the lexical entry for *dist1*. The VARIABLE value of the third elementary predication in the fifth set, the MEMBER value of the fourth elementary predication in the fifth set, and the OBJECT value of the sole elementary predication in the fourth set are required to be identical to each other primarily by the lexical

$$\begin{aligned}
& \left\langle \left\{ \begin{bmatrix} \text{HNDL} & h1 \\ \text{RELN} & \text{the} \\ \text{VARIABLE} & X \\ \text{RESTRICTOR} & h2 \\ \text{SCOPE} & h3 \end{bmatrix} \right\}, \left\{ \begin{bmatrix} \text{HNDL} & h4 \\ \text{RELN} & \text{cats} \\ \text{INSTANCE} & X \end{bmatrix} \right\}, \right. \\
& \left. \left\{ \begin{bmatrix} \text{HNDL} & h5 \\ \text{RELN} & \text{every} \\ \text{VARIABLE} & y \\ \text{RESTRICTOR} & h6 \\ \text{SCOPE} & h7 \end{bmatrix} \right\}, \left\{ \begin{bmatrix} \text{HNDL} & h6 \\ \text{RELN} & \text{member_of} \\ \text{MEMBER} & y \\ \text{SET} & X \end{bmatrix} \right\}, \left\{ \begin{bmatrix} \text{HNDL} & h8 \\ \text{RELN} & \text{saw} \\ \text{SUBJECT} & y \\ \text{OBJECT} & z \end{bmatrix} \right\}, \right. \\
& \left. \left\{ \begin{bmatrix} \text{HNDL} & h9 \\ \text{RELN} & \exists \\ \text{VARIABLE} & f \\ \text{SCOPE} & h10 \end{bmatrix} \right\}, \left\{ \begin{bmatrix} \text{HNDL} & h10 \\ \text{RELN} & \text{each_other} \\ \text{ARG1} & f \\ \text{ARG2} & h11 \end{bmatrix} \right\}, \right. \\
& \left. \left\{ \begin{bmatrix} \text{HNDL} & h12 \\ \text{RELN} & \text{every} \\ \text{VARIABLE} & z \\ \text{RESTRICTOR} & h13 \\ \text{SCOPE} & h14 \end{bmatrix} \right\}, \left\{ \begin{bmatrix} \text{HNDL} & h13 \\ \text{RELN} & \text{member_of} \\ \text{MEMBER} & z \\ \text{SET} & f(y) \end{bmatrix} \right\} \right\rangle
\end{aligned}$$

Figure 4: A semantic representation that can be assigned to (35)

entry for *each other*. The VARIABLE value of the first elementary predication in the fifth set, the ARG1 value of the second elementary predication in the fifth set, and the unparenthesized portion of the SET value of the fourth elementary predication in the fifth set are required to be identical to each other primarily by the lexical entry for *each other* as well. And the parenthesized portion of the SET value of the fourth elementary predication in the fifth set is required to be identical to the SUBJECT value y of the sole elementary predication in the fourth set by the binding condition for reciprocals stated in (32), because the elementary predication in the fourth set, i.e. the elementary predication contributed by the verb *saw*, is the only elementary predication in this semantic representation in which the MEMBER value z of the fourth elementary predication in the fifth set is outranked by some variable and z is outranked by y in that elementary predication.

There are some constraints on scope relations that come from the standard components of HPSG and MRS. The handle $h2$ has to be locationally equivalent to $h4$ because (i) the lexical entry for *the* requires $h2$ to either outscope or be locationally equivalent to $h4$ and (ii) the NP complement of the D *the* does not contain any scope-taking element. The variable-binding condition requires that the definite article *the* take scope over the universal quantifier expressed by the *every* predicate in the third set because the restrictor of the latter contains the variable X , which must be bound by the former. Likewise, the variable-binding condition requires that the universal quantifier in the third set take scope over the universal quantifier in the fifth set, because the restrictor of the latter contains the variable y , which must be bound by the former. And the variable-binding condition also requires that the existential quantifier in the fifth set take scope over the universal quantifier in the same set, because the restrictor of the latter contains the variable f , which must be bound by the former.

These constraints, however, do not resolve all the scope ambiguities present in the semantic representation. The existential quantifier contributed by the reciprocal pronoun may or may not

take scope over the quantifier contributed by the definite article *the* and the universal quantifier contributed by the silent distribution operator. That means that we still have the following three possibilities open.

- (36) $\text{the}(X, \text{cats}(X),$
 $\quad \text{every}(y, \text{member_of}(y, X),$
 $\quad \quad \exists f[\text{each_other}(f,$
 $\quad \quad \quad \text{every}(z, \text{member_of}(z, f(y)),$
 $\quad \quad \quad \text{saw}(y, z))])])$
- (37) $\text{the}(X, \text{cats}(X),$
 $\quad \exists f[\text{each_other}(f,$
 $\quad \quad \text{every}(y, \text{member_of}(y, X),$
 $\quad \quad \text{every}(z, \text{member_of}(z, f(y)),$
 $\quad \quad \text{saw}(y, z))])])$
- (38) $\exists f[\text{each_other}(f,$
 $\quad \text{the}(X, \text{cats}(X),$
 $\quad \quad \text{every}(y, \text{member_of}(y, X),$
 $\quad \quad \text{every}(z, \text{member_of}(z, f(y)),$
 $\quad \quad \text{saw}(y, z)))]$

I note at this point that the two conditions in (27) are both satisfied in each of the three semantic representations.

Let us consider these three possibilities one by one, starting with (36). Given the way the *each_other* predicate is defined in (11), the representation in (36) necessarily fails to have a denotation, for the following reason. Let g be the assignment under which the *each_other* predication (i.e. the portion of the semantic representation that consists of the *each_other* predicate and its two arguments) is interpreted. The second argument of the *each_other* predicate can have a denotation under g as long as the domain of $g(f)$ contains one entity, namely $g(y)$. Therefore, in order for the *each_other* predication to be allowed to have a denotation, the domain of $g(f)$ cannot contain more than one entity, because f is required to be appropriately minimal under g for the second argument of the *each_other* predicate. Since there is no reciprocal function whose domain does not contain more than one entity, this means that the denotation of the *each_other* predication here is necessarily undefined, preventing the entire representation from having a denotation.

The representation in (38) likewise necessarily fails to have a denotation, because whenever f is appropriately minimal under some assignment g for the second argument of the *each_other* predicate in the sense defined in (12), f is appropriately minimal under some assignment h such that $h \supseteq g$ for the *every* predication starting on line 3 (i.e. the portion of the representation consisting of the *every* predicate on line 3 and its three arguments), in violation of the third clause of (11a). Notice that the second argument of the *each_other* predicate is allowed to have a denotation on condition that the domain of the function denoted by f contains all the cats and that the *every* predication starting on line 3 is allowed to have a denotation on exactly the same condition.

In contrast, the representation in (37) has a denotation; its denotation is 1 if and only if there is a reciprocal function f such that each of the cats y saw every member of the set $f(y)$. Shown in (39) are some of the reciprocal functions that can be denoted by f when the set of cats being talked about is $\{A, B, C, D\}$.

- (39) a. $\{\langle A, \{B, C, D\} \rangle, \langle B, \{C, D, A\} \rangle, \langle C, \{D, A, B\} \rangle, \langle D, \{A, B, C\} \rangle\}$

- b. $\{\langle A, \{B\} \rangle, \langle B, \{A\} \rangle, \langle C, \{D\} \rangle, \langle D, \{C\} \rangle\}$
- c. $\{\langle A, \{B\} \rangle, \langle B, \{C\} \rangle, \langle C, \{D\} \rangle, \langle D, \{A\} \rangle\}$

The functions in (39a), (39b), and (39c) are meant to instantiate the schematizations that are referred to as Strong Reciprocity, Partitioned Strong Reciprocity, and Weak Reciprocity respectively in Langendoen (1978). The definitions of those three schematizations are given in (40).

- (40) a. Strong Reciprocity: $(\forall x, y \in A)(x \neq y \rightarrow xRy)$
- b. Partitioned Strong Reciprocity: $(\forall i, 1 \leq i \leq n)(\forall x, y \in A_i)(x \neq y \rightarrow xRy)$
- c. Weak Reciprocity: $(\forall x \in A)(\exists y, z \in A)(x \neq y \wedge x \neq z \wedge xRy \wedge zRx)$

As can be seen from the interpretation assigned to (37), the truth conditions that the proposed theory assigns to a sentence containing a reciprocal with a plural antecedent are always weaker than those corresponding to Strong Reciprocity when the group denoted by the antecedent contains more than two entities, and that might appear to be a shortcoming of the theory, but in fact it is not, *pace* Dalrymple et al. (1998). In Dalrymple et al. (1998), the authors state that the fact that the following example sounds contradictory shows that the meaning of Strong Reciprocity “does not come from pragmatic strengthening, but is rather a genuine part of the reciprocal’s conventional meaning”.

- (41) # House of Commons legislators refer to each other indirectly; the most senior one addresses the most junior one directly.

I dispute this interpretation of example (41); in my view, the following interpretation is at least as plausible. The first sentence in (41) describes a recurring event, rather than a single event that takes place at a particular point in time, so it is not unreasonable to regard the sentence as containing an unpronounced adverb of quantification whose meaning is similar to that of *always*. The meaning expressed by the sentence can thus be something along the lines of “When some group of House of Commons legislators communicate with each other, they always refer to each other indirectly”, since the restrictor of a quantifier like *always* is determined contextually (see Beaver and Clark (2003)). Given this possibility, the first sentence of (41) is expected to be false when there are any pair of legislators who communicate with each other but do not refer to each other indirectly, even if the meaning expressed by the reciprocal in the sentence is weaker than that of Strong Reciprocity.

The findings reported in Poortman et al. (2018) are also consistent with the hypothesis that reciprocal pronouns never inherently express the meaning of Strong Reciprocity. In Poortman et al. (2018), the authors report, among other things, that more than one third of the participants of their experiment judged a Dutch sentence resembling the English sentence *The girls know each other* to be true in a situation where each of the three people referred to by the grammatical subject was supposed to know only one of the other two. This finding arguably militates against the view, expressed in Fiengo and Lasnik (1973) and Haug and Dalrymple (2020), that the truth conditions of a sentence like *The girls know each other* are those corresponding to Strong Reciprocity.

Let me next describe how example (1), which contains a split-antecedent reciprocal, is assigned an appropriate semantic representation by the grammar. As has already been noted, the proposed theory analyzes this sentence as involving right-node raising of the expression *each other’s names* out of two clauses, *Tom shouted each other’s names* and *Mary cried each other’s names*, which are conjoined with each other. Shown in Figure 5 and Figure 6 are possible order domains (i.e. DOM values) that the first clause and the second clause can be associated with by the grammar respectively. The elementary predication whose HNDL value is *h6* and the elementary predication whose HNDL value is *h14* are exempt anaphors in the sense defined in (31) (because the semantic

$$\begin{aligned}
& \left\langle \left[\begin{array}{l} \text{PHON } \langle \text{Tom} \rangle \\ \text{SYNSEM|CONT} \end{array} \right] \left[\begin{array}{l} \text{EP} \left\langle \left\{ \left[\begin{array}{l} \text{HNDL } h0 \\ \text{RELN } \text{identical} \\ \text{ARG1 } t \\ \text{ARG2 } \text{Tom} \end{array} \right] \right\} \right\rangle \end{array} \right] \right], \\
& \left[\begin{array}{l} \text{PHON } \text{shouted} \\ \text{SYNSEM|CONT} \end{array} \right] \left[\begin{array}{l} \text{EP} \left\langle \left\{ \left[\begin{array}{l} \text{HNDL } h1 \\ \text{RELN } \text{shouted} \\ \text{SUBJECT } t \\ \text{OBJECT } x \end{array} \right] \right\} \right\rangle \end{array} \right], \\
& \left[\begin{array}{l} \text{PHON } \langle \langle \text{each other} \rangle, 's \rangle, \langle \text{names} \rangle \rangle \\ \text{SYNSEM|CONT} \end{array} \right] \left[\begin{array}{l} \text{EP} \left\langle \left\{ \left[\begin{array}{l} \text{HNDL } h2 \\ \text{RELN } \exists \\ \text{VARIABLE } f \\ \text{SCOPE } h3 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h3 \\ \text{RELN } \text{each_other} \\ \text{ARG1 } f \\ \text{ARG2 } h4 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h5 \\ \text{RELN } \text{every} \\ \text{VARIABLE } w \\ \text{RESTRICTOR } h6 \\ \text{SCOPE } h7 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h6 \\ \text{RELN } \text{member_of} \\ \text{MEMBER } w \\ \text{SET } f(\boxed{a}) \end{array} \right] \right\} \right\rangle, \\
\left\langle \left[\begin{array}{l} \text{HNDL } h8 \\ \text{RELN } \text{some} \\ \text{VARIABLE } x \\ \text{RESTRICTOR } h9 \\ \text{SCOPE } h10 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h11 \\ \text{RELN } \text{name_or_names} \\ \text{INSTANCE } x \\ \text{POSSESSOR } w \end{array} \right] \right\rangle \end{array} \right] \right] \rangle
\end{aligned}$$

Figure 5: The order domain of the first conjunct of (1)

representations in Figure 5 and Figure 6 do not contain any elementary predications in which w or y is outranked), and as a result, the parenthesized portions of their SET values are not subject to any purely syntactic constraints; in Figure 5 and Figure 6, the parenthesized portions of the SET values are thus represented as $\boxed{a}$ and $\boxed{b}$ in order to underscore the fact that these values are not determined by syntactic constraints. I am assuming here that the determiner *each other's* in this example is semantically an argument of the following noun *names* and that the morpheme *'s* contained in it contributes an existential quantifier to the semantic representation, although these are details that are orthogonal to the primary topic of this article and I will therefore not provide the details of such an analysis.

In the theory of right-node raising advocated in Yatabe and Tam (2021), the order domain of a coordinate structure to which right-node raising has applied can consist of (i) an unpronounced domain object which carries the meaning of either conjunction or disjunction, (ii) domain objects which are obtained by compacting the portion of each conjunct that has not been right-node-raised, and (iii) domain objects that correspond to expressions that have been right-node-raised. The order domain of sentence (1) as a whole can thus consist of four domain objects, namely an unpronounced domain object, a second domain object representing the part of the first conjunct that has not been right-node-raised, a third domain object representing the part of the second conjunct that has not been right-node-raised, and a fourth domain object that represents the right-node-raised expression *each other's names*, as in Figure 7.

Each domain object in this order domain has the EP value that it does for the following reasons. The EP value of the first domain object is dictated by the phrase-structure schema that licenses coordinate structures (see Yatabe (2012) for the details). The EP value of the second domain object is required to be the concatenation of the EP values of two domain objects, namely the one corresponding to *Tom* and the one corresponding to *shouted* (see Yatabe and Tam (2021) and the references cited there for the details). Likewise, the EP value of the third domain object is required to be the concatenation of the EP values of the domain object corresponding to *Mary* and the one corresponding to *cried*. And the EP value of the fourth domain object, the one representing the right-node-raised DP, is determined according to the procedure proposed in Subsection 4.2 above. Since the EP value of the domain object for this DP is a list consisting of three sets in both

$$\begin{aligned}
& \left\langle \left[\begin{array}{l} \text{PHON } \langle \text{Mary} \rangle \\ \text{SYNSEM|CONT} \left[\begin{array}{l} \text{EP } \left\langle \left\{ \left[\begin{array}{l} \text{HNDL } h0 \\ \text{RELN } \text{identical} \\ \text{ARG1 } m \\ \text{ARG2 } \text{Mary} \end{array} \right] \right\} \right\rangle \end{array} \right] \right] \right\rangle, \\
& \left[\begin{array}{l} \text{PHON } \text{cried} \\ \text{SYNSEM|CONT} \left[\begin{array}{l} \text{EP } \left\langle \left\{ \left[\begin{array}{l} \text{HNDL } h12 \\ \text{RELN } \text{cried} \\ \text{SUBJECT } m \\ \text{OBJECT } z \end{array} \right] \right\} \right\rangle \end{array} \right] \right] \right\rangle, \\
& \left[\begin{array}{l} \text{PHON } \langle \langle \langle \text{each other} \rangle, 's \rangle, \langle \text{names} \rangle \rangle \\ \text{SYNSEM|CONT} \left[\begin{array}{l} \text{EP } \left\langle \left\{ \left[\begin{array}{l} \text{HNDL } h2 \\ \text{RELN } \exists \\ \text{VARIABLE } f \\ \text{SCOPE } h3 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h3 \\ \text{RELN } \text{each_other} \\ \text{ARG1 } f \\ \text{ARG2 } h4 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h13 \\ \text{RELN } \text{every} \\ \text{VARIABLE } y \\ \text{RESTRICTOR } h14 \\ \text{SCOPE } h15 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h14 \\ \text{RELN } \text{member_of} \\ \text{MEMBER } y \\ \text{SET } f(\boxed{b}) \end{array} \right] \right\} \right\rangle, \\ \left\{ \left[\begin{array}{l} \text{HNDL } h16 \\ \text{RELN } \text{some} \\ \text{VARIABLE } z \\ \text{RESTRICTOR } h17 \\ \text{SCOPE } h18 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h19 \\ \text{RELN } \text{name_or_names} \\ \text{INSTANCE } z \\ \text{POSSESSOR } y \end{array} \right] \right\} \right\rangle \end{array} \right] \right] \right\rangle \rangle
\end{aligned}$$

Figure 6: The order domain of the second conjunct of (1)

conjuncts, the EP value of this fourth domain object, which is created by merging the domain object for the DP in the first conjunct and the domain object for the DP in the second conjunct, ends up also being a list consisting of three sets. The first set is the union of the set representing the meaning that the word *each other* is given in the first conjunct and the set representing the meaning it is given in the second conjunct. The second set is the union of the set representing the meaning that the morpheme *'s* is given in the first conjunct and the set representing the meaning it is given in the second conjunct. And the third set is the union of the set representing the meaning that the word *names* is given in the first conjunct and the set representing the meaning it is given in the second conjunct.

In Yatabe and Tam’s theory, the semantic representation for the sentence as a whole is obtained as the SYNSEM|CONT|EP value of the domain object that results from compacting the matrix node of the sentence. In the case of sentence (1), the EP value of the domain object that results from compacting the matrix node is the list shown in Figure 8, which is the concatenation of all the EP values in Figure 7. This EP value is a list consisting of eight sets, but it can naturally be viewed as a collection of 15 elementary predications, and is meant to be interpreted as such.

The standard components of HPSG and MRS ensure that $h9 \approx h11$ and $h17 \approx h19$. The phrase-structure schema for coordinate structures requires, indirectly, that $h21$ either outscope or be locationally equivalent to $h1$ and that $h22$ either outscope or be locationally equivalent to $h12$. And the variable-binding condition requires (i) that the existential quantifier contributed by *each other* (i.e. the existential quantifier whose label is $h2$) outscope the two universal quantifiers, (ii) that the two universal quantifiers (whose labels are $h5$ and $h13$) outscope the two existential quantifiers coming from *'s* (whose labels are $h8$ and $h16$) respectively, (iii) that $h8$ outscope $h1$, and (iv) that $h16$ outscope $h12$.

I hypothesize with Fox (2000, Section 2.3),⁶ Johnson (2009), and Sportiche (2024) that a quantifier contained in a conjunct or a disjunct can take scope over the coordinate structure as a whole only if each of the other conjuncts or disjuncts contains a pronoun bound by that quantifier. One possible way to formulate this hypothesis, which completely eliminates the remaining scope

⁶Fox attributes the hypothesis to Eddy Ruys’s 1993 Universiteit Utrecht dissertation, *The Scope of Indefinites*.

$$\begin{aligned}
& \left\langle \left[\begin{array}{l} \text{PHON } none \\ \text{SYNSEM|CONT } \left[\begin{array}{l} \text{EP } \left\langle \left\{ \left[\begin{array}{l} \text{HNDL } h20 \\ \text{RELN } and \\ \text{CONJUNCTS } \langle h21, h22 \rangle \end{array} \right] \right\} \right\rangle \end{array} \right] \right] \right\rangle, \\
& \left[\begin{array}{l} \text{PHON } \langle \langle \text{Tom} \rangle, \text{shouted} \rangle \\ \text{SYNSEM|CONT } \left[\begin{array}{l} \text{EP } \left\langle \left\{ \left[\begin{array}{l} \text{HNDL } h0 \\ \text{RELN } identical \\ \text{ARG1 } t \\ \text{ARG2 } Tom \end{array} \right] \right\}, \left\{ \left[\begin{array}{l} \text{HNDL } h1 \\ \text{RELN } shouted \\ \text{SUBJECT } t \\ \text{OBJECT } x \end{array} \right] \right\} \right\rangle \end{array} \right] \right] \right\rangle, \\
& \left[\begin{array}{l} \text{PHON } \langle \text{and}, \langle \text{Mary} \rangle, \text{cried} \rangle \\ \text{SYNSEM|CONT } \left[\begin{array}{l} \text{EP } \left\langle \left\{ \left[\begin{array}{l} \text{HNDL } h0 \\ \text{RELN } identical \\ \text{ARG1 } m \\ \text{ARG2 } Mary \end{array} \right] \right\}, \left\{ \left[\begin{array}{l} \text{HNDL } h12 \\ \text{RELN } cried \\ \text{SUBJECT } m \\ \text{OBJECT } z \end{array} \right] \right\} \right\rangle \end{array} \right] \right] \right\rangle, \\
& \left[\begin{array}{l} \text{PHON } \langle \langle \langle \text{each other} \rangle, 's \rangle, \langle \text{names} \rangle \rangle \\ \text{SYNSEM|CONT } \left[\begin{array}{l} \text{EP } \left\langle \left\{ \left[\begin{array}{l} \text{HNDL } h2 \\ \text{RELN } \exists \\ \text{VARIABLE } f \\ \text{SCOPE } h3 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h3 \\ \text{RELN } each_other \\ \text{ARG1 } f \\ \text{ARG2 } h4 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h5 \\ \text{RELN } every \\ \text{VARIABLE } w \\ \text{RESTRICTOR } h6 \\ \text{SCOPE } h7 \end{array} \right] \right\}, \right. \\ \left. \left\{ \left[\begin{array}{l} \text{HNDL } h6 \\ \text{RELN } member_of \\ \text{MEMBER } w \\ \text{SET } f(\boxed{a}) \end{array} \right], \left[\begin{array}{l} \text{HNDL } h13 \\ \text{RELN } every \\ \text{VARIABLE } y \\ \text{RESTRICTOR } h14 \\ \text{SCOPE } h15 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h14 \\ \text{RELN } member_of \\ \text{MEMBER } y \\ \text{SET } f(\boxed{b}) \end{array} \right] \right\}, \\ \left\{ \left[\begin{array}{l} \text{HNDL } h8 \\ \text{RELN } some \\ \text{VARIABLE } x \\ \text{RESTRICTOR } h9 \\ \text{SCOPE } h10 \end{array} \right], \left[\begin{array}{l} \text{HNDL } h16 \\ \text{RELN } some \\ \text{VARIABLE } z \\ \text{RESTRICTOR } h17 \\ \text{SCOPE } h18 \end{array} \right] \right\}, \\ \left\{ \left[\begin{array}{l} \text{HNDL } h11 \\ \text{RELN } name_or_names \\ \text{INSTANCE } x \\ \text{POSSESSOR } w \end{array} \right], \left[\begin{array}{l} \text{HNDL } h19 \\ \text{RELN } name_or_names \\ \text{INSTANCE } z \\ \text{POSSESSOR } y \end{array} \right] \right\} \right\rangle \end{array} \right] \right] \right\rangle \rangle
\end{aligned}$$

Figure 7: The order domain of sentence (1) as a whole

$$\begin{aligned}
& \left\langle \left[\begin{array}{ll} \text{HNDL} & h20 \\ \text{RELN} & \text{and} \\ \text{CONJUNCTS} & \langle h21, h22 \rangle \end{array} \right] \right\rangle, \left\{ \left[\begin{array}{ll} \text{HNDL} & h0 \\ \text{RELN} & \text{identical} \\ \text{ARG1} & t \\ \text{ARG2} & \text{Tom} \end{array} \right] \right\}, \left\{ \left[\begin{array}{ll} \text{HNDL} & h1 \\ \text{RELN} & \text{shouted} \\ \text{SUBJECT} & t \\ \text{OBJECT} & x \end{array} \right] \right\}, \\
& \left\{ \left[\begin{array}{ll} \text{HNDL} & h0 \\ \text{RELN} & \text{identical} \\ \text{ARG1} & m \\ \text{ARG2} & \text{Mary} \end{array} \right] \right\}, \left\{ \left[\begin{array}{ll} \text{HNDL} & h12 \\ \text{RELN} & \text{cried} \\ \text{SUBJECT} & m \\ \text{OBJECT} & z \end{array} \right] \right\}, \\
& \left\{ \left[\begin{array}{ll} \text{HNDL} & h2 \\ \text{RELN} & \exists \\ \text{VARIABLE} & f \\ \text{SCOPE} & h3 \end{array} \right], \left[\begin{array}{ll} \text{HNDL} & h3 \\ \text{RELN} & \text{each_other} \\ \text{ARG1} & f \\ \text{ARG2} & h4 \end{array} \right], \left[\begin{array}{ll} \text{HNDL} & h5 \\ \text{RELN} & \text{every} \\ \text{VARIABLE} & w \\ \text{RESTRICTOR} & h6 \\ \text{SCOPE} & h7 \end{array} \right], \right. \\
& \left. \left[\begin{array}{ll} \text{HNDL} & h6 \\ \text{RELN} & \text{member_of} \\ \text{MEMBER} & w \\ \text{SET} & f(\boxed{a}) \end{array} \right], \left[\begin{array}{ll} \text{HNDL} & h13 \\ \text{RELN} & \text{every} \\ \text{VARIABLE} & y \\ \text{RESTRICTOR} & h14 \\ \text{SCOPE} & h15 \end{array} \right], \left[\begin{array}{ll} \text{HNDL} & h14 \\ \text{RELN} & \text{member_of} \\ \text{MEMBER} & y \\ \text{SET} & f(\boxed{b}) \end{array} \right] \right\}, \\
& \left\{ \left[\begin{array}{ll} \text{HNDL} & h8 \\ \text{RELN} & \text{some} \\ \text{VARIABLE} & x \\ \text{RESTRICTOR} & h9 \\ \text{SCOPE} & h10 \end{array} \right], \left[\begin{array}{ll} \text{HNDL} & h16 \\ \text{RELN} & \text{some} \\ \text{VARIABLE} & z \\ \text{RESTRICTOR} & h17 \\ \text{SCOPE} & h18 \end{array} \right] \right\}, \\
& \left\{ \left[\begin{array}{ll} \text{HNDL} & h11 \\ \text{RELN} & \text{name_or_names} \\ \text{INSTANCE} & x \\ \text{POSSESSOR} & w \end{array} \right], \left[\begin{array}{ll} \text{HNDL} & h19 \\ \text{RELN} & \text{name_or_names} \\ \text{INSTANCE} & z \\ \text{POSSESSOR} & y \end{array} \right] \right\} \rangle
\end{aligned}$$

Figure 8: A possible semantic representation for sentence (1)

ambiguity in the present case, is shown in (42). The terms *to bind into* and *to non-pronominally bind into*, which are used in (42), are defined in (43). The reader is referred to Yatabe (2022) for an explanation about the ANTECEDENT feature, mentioned in (43).

- (42) When a quantifier non-pronominally binds into a conjunct or disjunct of a semantic coordinate structure, it must bind into all the other conjuncts or disjuncts of that semantic coordinate structure.
- (43) a. A quantifier that binds a variable x is said to *bind into* a constituent of a semantic representation if and only if (i) that constituent is part of the nuclear scope of that quantifier and (ii) x occurs in that constituent.
- b. A quantifier that binds a variable x is said to *non-pronominally bind into* a constituent of a semantic representation if and only if (i) that constituent is part of the nuclear scope of that quantifier and (ii) x occurs in that constituent as the value of, and not as a proper part of the value of, a feature other than ANTECEDENT.

Given the constraint in (42), neither the universal quantifier labeled by $h5$ nor the existential quantifier labeled by $h8$ can outscope the conjunction in the semantic representation shown in Figure 8; if only one of these quantifiers (which has to be the universal quantifier because it is required by the variable-binding condition to outscope the existential quantifier) or both of the quantifiers outscope the conjunction, then there will be a quantifier that non-pronominally binds into the first semantic conjunct without binding into the second semantic conjunct. Likewise, neither the universal quantifier labeled by $h13$ nor the existential quantifier labeled by $h16$ can outscope the conjunction; if only one of these quantifiers (which has to be the universal quantifier) or both of them outscope the conjunction, then there will be a quantifier that non-pronominally binds into the second semantic conjunct without binding into the first. This leaves us with the semantic representation shown in (44). Note that the predicate *identical*, which I introduced in Subsection 4.1 above, will henceforth be represented by an infixed equal sign.

- (44) $t = \text{Tom} \wedge m = \text{Mary}$
 $\wedge \exists f[\text{each_other}(f,$
 $\quad \text{and}(\text{every}(w, \text{member_of}(w, f(\boxed{a})),$
 $\quad \quad \text{some}(x, \text{name_or_names_of}(x, w),$
 $\quad \quad \text{shouted}(t, x))),$
 $\quad \text{every}(y, \text{member_of}(y, f(\boxed{b})),$
 $\quad \quad \text{some}(z, \text{name_or_names_of}(z, y),$
 $\quad \quad \text{cried}(m, z)))))]$

The conditions stated in (27), which are meant to be part of the lexical entry for *each other*, require that the denotation of $\boxed{a}$ not be identical to w and that the nuclear scope of the first universal quantifier contain an occurrence of the variable denoted by $\boxed{a}$. Since the nuclear scope of the first universal quantifier contains three variables, namely t , w , and x , and the variable-binding condition prevents the denotation of $\boxed{a}$ from being x , it follows that the denotation of $\boxed{a}$ must be t . And since the denotation of $\boxed{b}$ must be m for similar reasons, (13) is obtained as a semantic representation for sentence (1).

I have just derived one possible semantic representation of sentence (1) assuming that the upper portions of reciprocal meaning coming from the two conjuncts happened to be identical to each other and that the lower portions coming from the two conjuncts happened not to be identical to each other. Let us next examine what happens when that assumption is dropped. In doing so, I will refer to the four elementary predications shown in Figure 3 as the first, the second, the third, and the fourth elementary predication, from left to right.

When the first elementary predications coming from the two conjuncts happen to be identical to each other but the second elementary predications coming from the two conjuncts happen not to be, there will be no way to integrate all the elementary predications into a well-formed MRS representation.⁷ Owing to the variable-binding condition, we likewise fail to obtain a well-formed MRS representation (i) when the second elementary predications coming from the two conjuncts happen to be identical but the first elementary predications coming from the two conjuncts happen not to be and (ii) when the fourth elementary predications coming from the two conjuncts happen to be identical but the third elementary predications coming from the two conjuncts happen not to be.

When the third elementary predications coming from the two conjuncts happen to be identical to each other but the fourth elementary predications coming from the two conjuncts happen not to be, we do obtain a well-formed semantic representation, but a semantic representation obtained in such a way necessarily fails to have a denotation because it contains a formula of the following form.

$$(45) \quad \text{every}(w, \text{member_of}(w, f(\boxed{a})) \wedge \text{member_of}(w, f(\boxed{b})), \dots)$$

First, suppose the denotations of $\boxed{a}$ and $\boxed{b}$ are the same variable. In that case, an *each_other* predication will fail to have a denotation because its second argument is allowed to have a denotation even if the domain of the function denoted by f contains only one entity (and there is no reciprocal function whose domain contains only one entity). Next, suppose the denotations of $\boxed{a}$ and $\boxed{b}$ are different variables. In that case, the *every* predication shown in (45) fails to have a denotation, because f will have to be mapped to a reciprocal function whose domain contains the denotations of $\boxed{a}$ and $\boxed{b}$ and nothing else, which entails that no entity can satisfy the condition specified by the restrictor of *every* (i.e. its second argument).

When the lower portions of reciprocal meaning coming from the two conjuncts happen to be identical to each other but the upper portions coming from the two conjuncts happen not to be identical (in the sense that the first elementary predications coming from the two conjuncts happen not to be identical and the second elementary predications coming from the two conjuncts also happen not to be identical), we fail to obtain a well-formed semantic representation again, due to the variable-binding condition.

When the upper portions of reciprocal meaning coming from the two conjuncts happen to be identical to each other and the lower portions coming from the two conjuncts also happen to be identical to each other, we obtain a well-formed semantic representation, but a semantic representation obtained in that way cannot have a denotation either, because in such a semantic representation, the second argument of the *each_other* predicate contains only one formula of the form $f(x)$, where f is a variable forced to denote a reciprocal function and x is a variable

⁷This can be shown as follows. First, suppose the conjunction expressed by the *and* predicate is outscoped by the existential quantifier binding the variable that is forced to represent a reciprocal function. In that case, at least one of the two *each_other* predicates will neither outscope nor be outscoped by the conjunction, and there will consequently be no elementary predication that can serve as the second argument of that *each_other* predicate, because (i) the elementary predications contributed by the verbs *shouted* and *cried* are both required (by the phrase-structure schema for coordinate structures) to be outscoped by the conjunction, (ii) the existential quantifier or quantifiers contributed by 's are required (by the variable-binding condition) to outscope one of the elementary predications coming from the verbs, and (iii) the universal quantifier or quantifiers coming from the reciprocal are required (by the variable-binding condition) to outscope an existential quantifier coming from 's. Next, suppose the conjunction is not outscoped by the existential quantifier binding the variable that is forced to represent a reciprocal function. In that case, at least one of the conjuncts will neither outscope nor be outscoped by that existential quantifier, and the variable-binding condition will consequently be violated because one occurrence of the variable that is to be bound by that existential quantifier is contained in the restrictor of an existential quantifier assumed to be contributed by 's, a quantifier that has to either outscope or be outscoped by the conjunct in question.

that cannot be bound by a universal quantifier (due to the constraint in (27a)), creating a situation where f is forced to denote a reciprocal function whose domain contains only one entity, a function that does not exist.

And when the upper portions of reciprocal meaning coming from the two conjuncts happen not to be identical to each other (in the sense that the first elementary predications coming from the two conjuncts happen not to be identical and the second elementary predications coming from the two conjuncts also happen not to be identical) and the lower portions coming from the two conjuncts also happen not to be identical to each other (in the sense that the third elementary predications coming from the two conjuncts happen not to be identical and the fourth elementary predications coming from the two conjuncts also happen not to be identical), we obtain a well-formed semantic representation containing two existentially bound variables that are forced to denote reciprocal functions, but both the variables are required to denote a reciprocal function whose domain contains only one entity, preventing the semantic representation from having a denotation.

At this point, all the possibilities to consider have been considered, and it has been ascertained that the proposed theory does not yield any spurious reading for sentence (1).

5 The scope of reciprocals

In Section 3 above, I noted that the third clause of (11a) is meant to prevent the *each_other* predicate from taking excessively wide scope. I reproduce (11a) in its entirety here as (46).

- (46) $\llbracket \text{each_other}(f, X) \rrbracket^{M,g}$ is defined and is either 1 or 0 if and only if
- $g(f)$ is a reciprocal function,
 - f is appropriately minimal for X under g , and
 - X contains no sub-constituent for which f is appropriately minimal under some assignment h such that $h \supseteq g$.

Inclusion of the third clause in this definition is motivated by examples like (47), which I owe to Daisuke Bekki, and (48).

- (47) It is not true that the cats saw each other.
(48) I didn't know that the cats knew each other.

If the *each_other* predicate were allowed to take scope over the matrix predicate in a semantic representation for (47), the sentence could be associated with a semantic representation like (49) and thus would be predicted to be able to mean “There is a reciprocal function f such that it is not true that each cat x in a certain set of cats saw every member of the set $f(x)$ ”.

- (49) $\exists f [\text{each_other}(f,$
 $\quad \text{not}(\text{true}(\text{the}(X, \text{cats}(X),$
 $\quad \quad \text{every}(y, \text{member_of}(y, X),$
 $\quad \quad \quad \text{every}(z, \text{member_of}(z, f(y)),$
 $\quad \quad \quad \quad \text{saw}(y, z)))))))]$

That would be an incorrect prediction; in a situation where cat A saw cat B, cat B saw cat C, and cat C saw cat A, this sentence cannot be truthfully used, even if it is simultaneously the case that cat A did not see cat C, cat B did not see cat A, and cat C did not see cat B. The third clause of (46) prevents a semantic representation like (49) from having a denotation and thus forces the

each_other predicate in this sentence to take scope within the embedded clause, as in (50), giving the sentence as a whole an appropriate meaning to the effect that “It is not true that there is a reciprocal function f such that each cat x in a certain set of cats saw every member of the set $f(x)$ ”.

- (50) $\text{not}(\text{true}(\text{the}(X, \text{cats}(X),$
 $\quad \exists f[\text{each_other}(f,$
 $\quad \quad \text{every}(y, \text{member_of}(y, X),$
 $\quad \quad \quad \text{every}(z, \text{member_of}(z, f(y)),$
 $\quad \quad \quad \quad \text{saw}(y, z))))))$

Similarly, if it were not for the third clause of (46), the sentence in (48) would be incorrectly predicted to be able to mean “There is a reciprocal function f such that I didn’t know that each cat x knew every member of $f(x)$ ”. Because of the third clause, it is correctly predicted that the sentence can only mean “I didn’t know that there was a reciprocal function f such that each cat x knew every member of $f(x)$ ”.

This restriction on the scope of the *each_other* predicate does not interfere with the theory’s ability to assign adequate truth conditions to sentences like (51), discussed extensively in Heim, Lasnik, and May (1991a) and Heim, Lasnik, and May (1991b).

- (51) John and Mary think they like each other.

This sentence can mean “John and Mary think they (that is, John and Mary) like each other”, and it can also mean “John thinks he likes Mary, and Mary thinks she likes John”. In the theory proposed here, this fact receives an explanation in terms of scope ambiguity, as in the theory proposed in Heim, Lasnik, and May (1991b): the sentence expresses the first meaning when an unpronounced distribution operator is associated with the embedded subject and the *each_other* predicate takes scope immediately above it, and it expresses the second meaning when an unpronounced distribution operator is associated with the matrix subject and the *each_other* predicate takes scope immediately above it. It is obvious that the third clause of (46) does not prevent the sentence from receiving the first interpretation, which is induced by the reciprocal taking narrow scope in relation to the propositional attitude verb. In addition, it can be ascertained that the third clause does not prevent the sentence from receiving the second interpretation either. One semantic representation that can be assigned to the sentence in order to induce its second interpretation is (52).

- (52) $j = \text{John} \wedge m = \text{Mary}$
 $\wedge \text{some}(X, X = \{j, m\},$
 $\quad \exists f[\text{each_other}(f,$
 $\quad \quad \text{every}(y, y \in X,$
 $\quad \quad \quad \text{think}(y, \text{some}(z, z = y,$
 $\quad \quad \quad \quad \text{every}(w, \text{member_of}(w, f(z)),$
 $\quad \quad \quad \quad \quad \text{like}(z, w))))))])$

This semantic representation is true if and only if the *each_other* predicate is satisfied under an assignment that maps f to the reciprocal function whose domain is {John, Mary}. Notice that the second argument of the *each_other* predicate does not contain a sub-constituent that is allowed to have a denotation only if the domain of the function denoted by f contains both John and Mary; the third argument of the *every* predicate, for instance, has a denotation as long as the domain of the function denoted by f contains one entity, namely whatever value is assigned to y , to which z , the argument of f , is required to be identical.

The account just described is arguably superior to the account proposed in Heim, Lasnik, and May (1991a) and Heim, Lasnik, and May (1991b) in the following respect. It is pointed out in Dimitriadis (2000) that the sentence in (53) can mean “John thinks his mother likes Mary’s mother, and Mary thinks her mother likes John’s mother”. This putative fact is unexpected in Heim et al.’s account.

(53) John and Mary think their mothers like each other. (Dimitriadis, 2000)

In the theory proposed in Heim, Lasnik, and May (1991b), the matrix subject of this sentence must be viewed as the semantic antecedent of the reciprocal, i.e. a plural DP that is a sister of the silent distribution operator that binds one of the indices associated with the reciprocal. Since the binding condition for reciprocals requires the embedded subject, *their mothers*, to be the syntactic antecedent of the reciprocal, it turns out that the syntactic antecedent and the semantic antecedent of the reciprocal are not coindexed in this sentence, which is supposed to be an impossibility in their theory. Simply allowing the syntactic and the semantic antecedent of a reciprocal not to be coindexed does not solve the problem, as the theory thus modified will incorrectly predict that the sentence can mean something like “Each member x of the set {John, Mary} thinks that x ’s mother likes every member of the set {John, Mary} other than x ”, that is to say, “John thinks his mother likes Mary, and Mary thinks her mother likes John”. In contrast, in the theory proposed in this article, the sentence receives the following semantic representation, which expresses appropriate truth conditions, without violating the binding condition for reciprocals or any other grammatical constraints.

(54) $j = \text{John} \wedge m = \text{Mary}$
 $\wedge \text{some}(X, X = \{j, m\},$
 $\quad \exists f[\text{each_other}(f,$
 $\quad \quad \text{every}(y, y \in X,$
 $\quad \quad \quad \text{think}(y, \text{some}(z, z = y,$
 $\quad \quad \quad \quad \text{some}(w, \text{mother_or_mothers_of}(w, z),$
 $\quad \quad \quad \quad \quad \text{every}(v, \text{member_of}(v, f(w)),$
 $\quad \quad \quad \quad \quad \quad \text{like}(w, v)))))))]]$

In some cases, a reciprocal pronoun is prevented from taking wide scope not because of the third clause of (46), but because the reciprocal is contained in a scope island. One such case is (55), which, according to Heim, Lasnik, and May (1991a), has only “the contradictory, or narrow, reading”; it cannot mean “John criticized Max when he defeated Mary, and Mary criticized Max when she defeated John”.

(55) John and Mary criticized Max when they defeated each other.

As noted in Heim, Lasnik, and May (1991a), the inability of the reciprocal to take wide scope in this sentence can be understood as a consequence of the fact that the tensed adjunct clause in it is a scope island. In the theory advocated in Yatabe and Tam (2021), a phrase that is required to undergo an operation called total compaction becomes a scope island. The fact at hand can therefore be captured by adding to the grammar a constraint that requires tensed adjunct clauses to be totally compacted. Such a constraint will force the existential quantifier contributed by the reciprocal to take scope within the tensed adjunct clause.⁸

⁸It has been noted in the literature (see Barker (2022) and the references cited there) that a tensed adjunct clause does not function as a scope island for indefinite expressions such as *a relative of mine*. It needs to be assumed that existential quantifiers contributed by reciprocals are different from existential quantifiers contributed by indefinite expressions in this regard.

6 Simplification of the theory of right-node raising

In this article, I have proposed some modifications to the theory of left- and right-node raising presented in Yatabe and Tam (2021) in order to deal with split-antecedent reciprocals. In this penultimate section, I will point out that the proposed modifications lead to a simplification of that theory. Although the theory in question deals with both left- and right-node raising, I will focus on right-node raising below.

In Yatabe and Tam's theory, right-node raising (RNR) is divided into two types, namely a purely phonological, meaning-preserving type and a syntactic, potentially meaning-changing type, and the latter is further divided into three types. In syntactic RNR of the first type, completely identical domain objects are right-node-raised out of multiple phrases and fused into a single domain object. When the right-node-raised expression is a quantifier, it is interpreted as a single quantifier taking wide scope over the phrases out of which it has been right-node-raised. In syntactic RNR of the second type, a new domain object is created by fusing the domain objects that have been right-node-raised out of multiple phrases, and the newly created domain object retains inside it the meaning of each domain object that existed before the application of RNR. When the right-node-raised expression is a quantifier, it is interpreted as multiple quantifiers. In the case of the sentence in (3a), reproduced here as (56), the reading in which the phrase *every saxophonist* takes scope over the entire sentence is taken to result from the first type of syntactic RNR, and the readings that can be expressed by sentence (3b) as well are taken to result from the second type of syntactic RNR (or from phonological RNR, although that is an irrelevant possibility in the present context).

(56) Some woman likes and some man detests every saxophonist.

In syntactic RNR of the third type, which can be applied to conjunctive coordinate structures and to non-coordinate structures but not to disjunctive coordinate structures, a new domain object is created by fusing the domain objects that have been right-node-raised out of multiple phrases, as in the second type of syntactic RNR, and the newly created domain object is made to refer to the sum of the things that the domain objects that existed before the application of RNR referred to. This third type of syntactic RNR is assumed to be involved in a sentence like (4a), reproduced here as (57).

(57) John avoided and Bill ignored similar issues.

The theory assigns to this sentence an interpretation to the effect that there are some mutually similar issues $x + y$ such that John avoided x and Bill ignored y .

The method for fusing multiple domain objects that was proposed in Subsection 4.2 above subsumes both the first type and the second type of syntactic RNR, thus making it unnecessary to distinguish those two types of syntactic RNR. When the domain objects that have been right-node-raised out of multiple phrases share the same EP value, the EP value of the new domain object that is created by fusing them according to the proposed method will be identical to the EP value of each of those original domain objects, accounting for cases that were accounted for as instances of the first type of syntactic RNR in Yatabe and Tam (2021). In contrast, when the EP values of the domain objects that have been right-node-raised out of multiple phrases are entirely distinct from each other (in the sense that no elementary predication is contained in two or more sets representing the meaning of a single lexical item), the EP value of the new domain object that is created by fusing them according to the proposed method will retain all the elementary predications contained in the EP values of the original domain objects, accounting for cases that were accounted for as instances of the second type of syntactic RNR in Yatabe and Tam's theory.

In this new theory, a sentence like (1) represents an intermediate case, in which the domain objects that were right-node-raised out of two conjuncts had EP values that were partially identical.

7 Conclusion

I have argued in this article that the existence of split-antecedent reciprocals necessitates a new theory of reciprocals in which the meaning of a reciprocal involves an existentially bound variable representing a certain type of function, and proposed some modifications to the analysis of right-node raising proposed in Yatabe and Tam (2021) that allow this theory to be formulated in a straightforward fashion. The proposed theory is representationalist in the sense that it crucially relies on the view that the meaning of each lexical item is represented as a set of elementary predications and that semantic composition may involve taking the union of such sets. It is an open question whether it is possible to account for the same range of phenomena using a non-representationalist theory, which assumes that linguistic expressions are linked to their denotations directly without mediation through semantic representations.

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